

Math 8 Mid-Quarter Test

Mary, Star of the Sea School · Dr. Park · Q1–Q2 Content

Name: _____ Class: _____ Date: _____

Directions: Answer all 20 questions in the space provided. Show your work where helpful.

1. Write 82,500,000 in scientific notation.

Answer: _____

2. Write 0.00051 in scientific notation.

Answer: _____

3. Compute: $(4 \times 10^5)(3 \times 10^2)$. Give in scientific notation.

Answer: _____

4. Compute: $(9 \times 10^8) \div (3 \times 10^3)$. Give in scientific notation.

Answer: _____

5. Simplify: $3^4 \cdot 3^2$.

Answer: _____

6. Simplify: $(7^5) \div (7^2)$.

Answer: _____

7. Evaluate: $\sqrt{81}$.

Answer: _____

8. Evaluate: $\sqrt{144}$.

Answer: _____

9. Is $5/8$ rational or irrational?

Answer: _____

10. Find the slope of the line through (2, 3) and (5, 9).

Answer: _____

11. Find the slope of the line through (0, 6) and (3, 0).

Answer: _____

12. What is the y-intercept of $y = -2x + 5$?

Answer: _____

13. What is the slope of $y = (1/2)x - 4$?

Answer: _____

14. Solve for x: $6x - 5 = 19$.

Answer: _____

15. Solve for x : $3(x - 4) = 15$.

Answer: _____

16. Solve for x : $4x + 8 = 6x - 2$.

Answer: _____

17. A line has slope -3 and y -intercept 4 . Write it in slope-intercept form.

Answer: _____

18. At what x -value do $y = 2x + 1$ and $y = 4x - 5$ intersect?

Answer: _____

19. A right triangle has legs 6 and 8 . Find the hypotenuse.

Answer: _____

20. Convert 3.2×10^{-2} to standard form.

Answer: _____